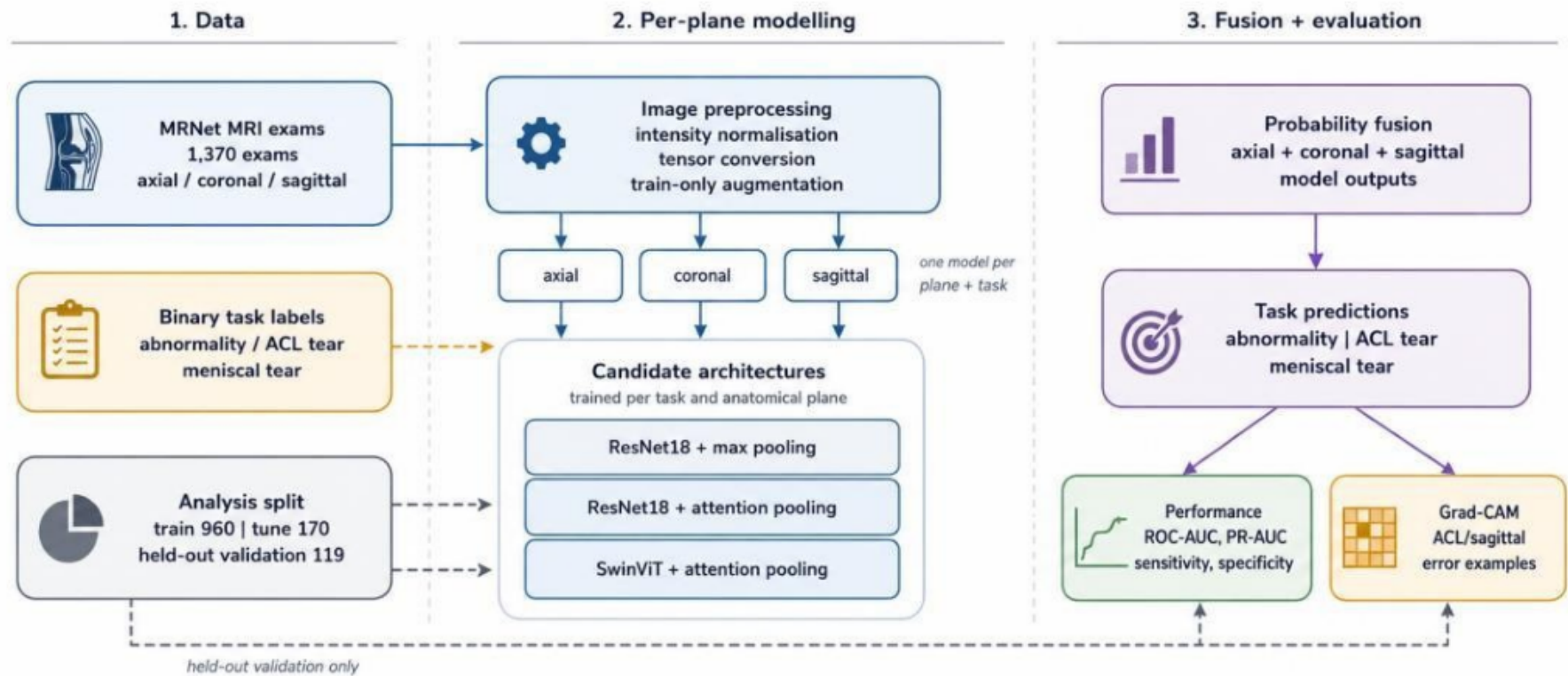


MRNet Knee MRI · Deep Learning Architecture

Per-Plane Modelling Workflow + Late Multi-Plane Probability Fusion

MRNet modelling pipeline

Three-plane MRI exams are processed per plane, fused at probability level, and evaluated on held-out validation data



Baseline ResNet-18

Slice Max-Pooling

Best overall aggregate (ROC-AUC 0.898)
Top ACL Tear detector (AUC 0.968)

Enhanced ResNet-18

Learned Attention

Two-stage trunk/gate optimization
Highest aggregate specificity (76.3%)

Swin Transformer

Shifted-Window Attn

Best Abnormality detection (AUC 0.928)
Top Meniscal Tear detector (AUC 0.804)

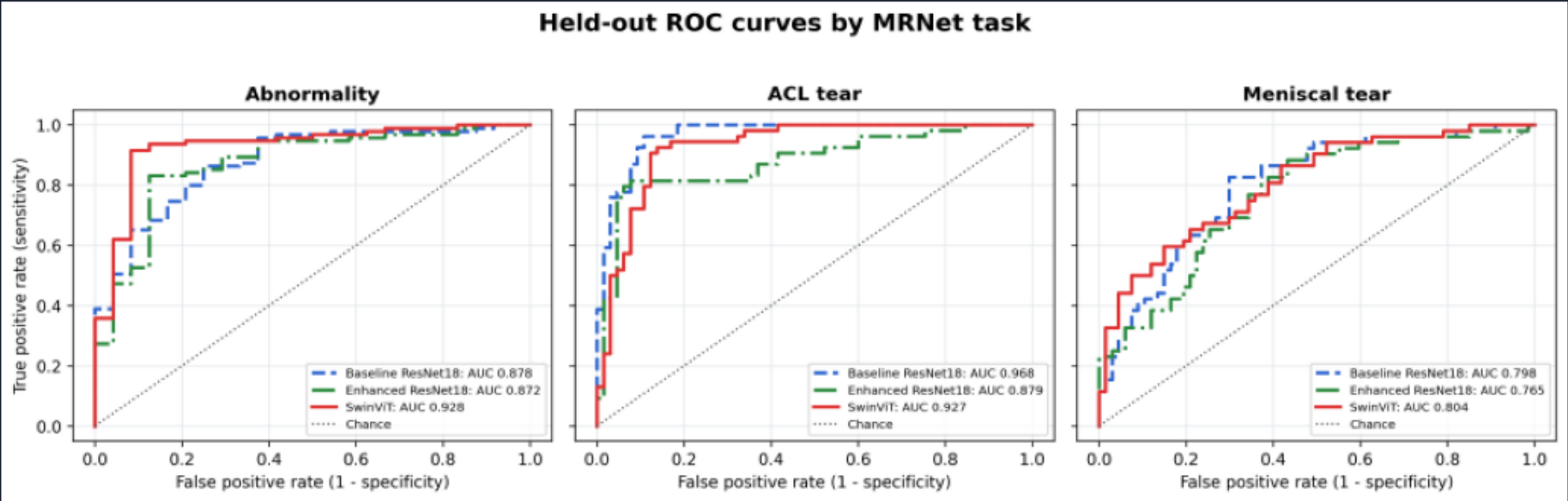
MRNet Knee MRI · Evaluation Benchmarks

Held-Out Test Set Performance (n=119 examinations, N=357 task predictions)

AGGREGATE BENCHMARK PERFORMANCE

Model Architecture	ROC-AUC	PR-AUC	Sensitivity	Specificity	F1 Score
Baseline ResNet-18 Fusion	0.898	0.910	90.5%	75.0%	0.863
SwinViT Fusion	0.887	0.907	88.1%	73.1%	0.843
Enhanced ResNet-18 (Attention)	0.859	0.890	81.6%	76.3%	0.816

HELD-OUT ROC CURVES BY TASK



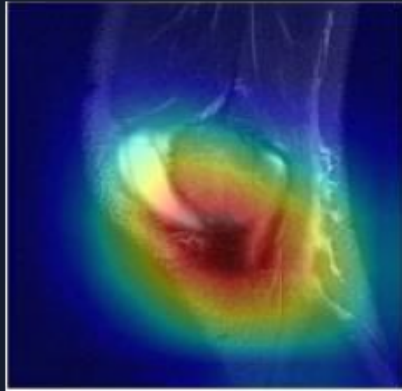
MRNet Knee MRI · Explainable AI

Grad-CAM Saliency Maps — Sagittal ACL Tear Error Analysis

TRUE POSITIVE

ID 0471 (Slice 4/33)

GT=1 · Pred=1 · $P > 0.999$

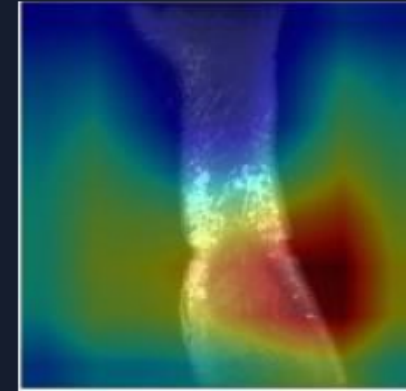


Focal, clinically valid attention
directly on anatomical ACL ligament

FALSE POSITIVE

ID 0544 (Slice 1/31)

GT=0 · Pred=1 · $P > 0.999$

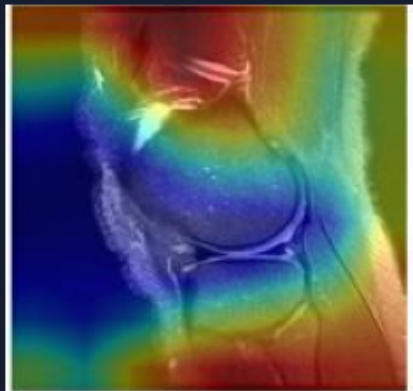


High confidence shortcut:
focus on posterior subcutaneous fat

FALSE NEGATIVE

ID 0969 (Slice 29/39)

GT=1 · Pred=0 · $P < 0.001$

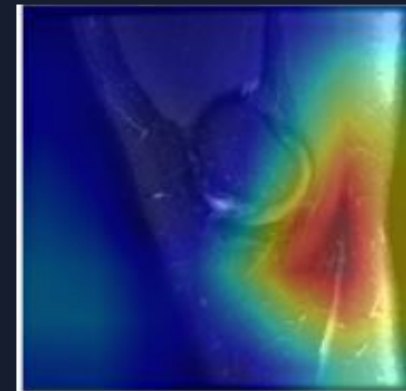


Subtle tear missed:
diffuse femur focus without ACL signal

TRUE NEGATIVE

ID 0023 (Slice 29/32)

GT=0 · Pred=0 · $P < 0.001$



Correct suppression:
clean negative across normal anatomy